

Module 16: Reproductive System

Topics: Male Reproductive System, Female Reproductive System, Pregnancy A&P (Basics)

TOPIC 1 OF 3 . WEEK 8 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Male Reproductive System

Testis structure, spermatogenesis, accessory glands, and the hormonal axis.

The Science

Testes do two jobs: produce sperm and produce testosterone. Sperm production happens in the seminiferous tubules, which are lined with Sertoli cells that support developing spermatogonia through meiosis. Testosterone is produced by Leydig cells (interstitial cells) outside the tubules. The blood-testis barrier (formed by Sertoli cell tight junctions) protects developing sperm from immune attack.

Sperm pathway: seminiferous tubule, rete testis, epididymis (sperm matures here, becomes motile, is stored), vas deferens (ascending through the inguinal canal), ejaculatory duct (joining with seminal vesicle duct), prostatic urethra, urethra. Accessory glands add to the seminal fluid: seminal vesicles (fructose, prostaglandins, about 60% of volume), prostate (alkaline secretion, enzymes, about 30%), bulbourethral glands (mucus pre-ejaculate).

Hormonal axis: hypothalamic GnRH drives pituitary LH and FSH. LH stimulates Leydig cells to make testosterone. FSH plus testosterone supports Sertoli cells and spermatogenesis. Testosterone feeds back negatively on the hypothalamus and pituitary to stabilize the system. Inhibin from Sertoli cells specifically suppresses FSH, fine-tuning spermatogenesis.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a cross-section of a testis with seminiferous tubules and Leydig cells labeled, then trace the path of sperm from tubule to urethra.

Common misconceptions to address

- Students think the testis is one structure with one function. It does sperm production AND testosterone production via different cells.
- The epididymis is sometimes overlooked. Sperm leaving the seminiferous tubules are not yet motile; they mature in the epididymis.
- Testosterone is thought of as 'the male hormone' but it is also made (in smaller amounts) by the adrenal cortex in both sexes.

Order of operations (what to teach first)

Testis structure and its two jobs, then sperm pathway with accessory glands, then the hormonal axis.

Self-test prompts

- Which cells produce testosterone?
- Which cells support developing sperm?
- Why do anabolic steroid users develop testicular atrophy?

Why This Matters to Your Patients

Testicular torsion is a surgical emergency; twisting of the spermatic cord cuts off blood supply and the testis dies within hours. Varicocele (enlarged pampiniform plexus) is a common cause of infertility because the heat impairs spermatogenesis. Benign prostatic hyperplasia (BPH) is age-related prostate enlargement causing urinary symptoms. Prostate cancer is the most common solid tumor in men; PSA screening is controversial but widely used. Hypogonadism (from various causes) produces low testosterone, low energy, reduced libido, reduced muscle mass; replacement therapy is increasingly common but has risks.

TOPIC 2 OF 3 . WEEK 8 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Female Reproductive System

Ovary, oogenesis, the ovarian and uterine cycles, and the hormonal choreography.

The Science

Ovaries produce eggs (oocytes) and hormones. Females are born with all their primordial follicles already arrested in prophase I of meiosis. Each cycle a cohort of follicles begins maturing; usually one becomes dominant. The dominant follicle's granulosa and theca cells produce estrogen under FSH and LH stimulation.

Ovarian cycle (about 28 days). Follicular phase (days 1-14): follicle matures; estrogen rises. Ovulation (day 14): LH surge triggers release of the secondary oocyte. Luteal phase (days 15-28): ruptured follicle becomes the corpus luteum, which produces progesterone (and some estrogen) to support potential implantation. If no pregnancy, the corpus luteum regresses, hormones drop, and the cycle restarts.

Uterine (menstrual) cycle parallels the ovarian: menstrual phase (days 1-5, shedding of endometrium), proliferative phase (days 6-14, estrogen rebuilds endometrium), secretory phase (days 15-28, progesterone makes endometrium glandular and receptive). If no implantation, progesterone falls and the cycle restarts.

Hormonal control: GnRH from hypothalamus drives LH and FSH. Estrogen and progesterone feed back, mostly negatively. The key twist: rising estrogen mid-cycle switches feedback from negative to positive, triggering the LH surge. After ovulation, the corpus luteum's progesterone restores negative feedback. If pregnancy occurs, the embryo's hCG keeps the corpus luteum alive until the placenta takes over hormone production.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the ovarian cycle phases with FSH, LH, estrogen, and progesterone curves over 28 days. Then label the uterine phases underneath.

Common misconceptions to address

- Students think eggs are made monthly. All eggs are present from birth; the monthly cycle is which one matures.
- Estrogen is thought to only feed back negatively. The mid-cycle switch to positive feedback is the key trigger for ovulation.
- The menstrual cycle is reduced to the bleeding phase. It is the integration of ovarian and uterine cycles over the entire month.

Order of operations (what to teach first)

Ovarian anatomy and oogenesis, then the ovarian cycle, then the uterine cycle in parallel, then hormonal control with the feedback switch.

Self-test prompts

- What triggers ovulation?
- What does the corpus luteum produce?
- How do combined oral contraceptives prevent pregnancy?

Why This Matters to Your Patients

Polycystic ovary syndrome (PCOS): irregular cycles, anovulation, hyperandrogenism, insulin resistance; common cause of infertility. Endometriosis: endometrial tissue outside the uterus producing painful, cyclic bleeding and infertility. Menopause: ovarian estrogen production declines; hot flashes, bone loss, mood changes. Hormonal contraception, fertility treatment, and management of menstrual disorders all depend on understanding the axis. Pap smears, mammograms, and pelvic exams are screening tools nurses help administer. Pregnancy testing detects hCG.

TOPIC 3 OF 3 . WEEK 8 . 7 CARDS (4 DOK 1 / 2 DOK 2 / 1 DOK 3)

Pregnancy A&P (Basics)

Fertilization, implantation, and the maternal-fetal interface. Maternal physiology highlights through the trimesters.

The Science

Fertilization happens in the ampulla of the uterine tube within hours of ovulation. The zygote divides as it travels toward the uterus and arrives as a blastocyst. Implantation in the endometrium is around day 6-7. Trophoblast cells secrete hCG almost immediately; hCG keeps the corpus luteum alive so it continues producing the progesterone that holds the endometrium in place. By the end of the first trimester, the placenta takes over hormone production.

The placenta is the working interface between mother and fetus. It exchanges gases, nutrients, and waste, but maternal and fetal blood never directly mix. It is also an endocrine organ (progesterone, estrogen, hPL) and an imperfect filter (many drugs, alcohol, and infections cross to the fetus).

Maternal physiology changes substantially. Blood volume rises about 40 percent by the second trimester. Cardiac output increases. Renal filtration goes up. The respiratory drive shifts because progesterone affects ventilation. Late in pregnancy, the enlarged uterus presses on the inferior vena cava when the mother lies supine, compressing venous return and dropping blood pressure: classic supine hypotensive syndrome of pregnancy. For essentials scope, skip the developmental biology details (germ layers, organogenesis week-by-week) and focus on the maternal-fetal interface.

Teaching This Topic

Before the video (drawing prompt)

Ask students to draw the maternal-fetal interface: maternal blood on one side, fetal vessels on the other, the placental membrane between. They have to commit to whether blood mixes (it does not) before the video confirms.

Common misconceptions to address

- Maternal and fetal blood mix at the placenta. They do not; only molecules diffuse across the placental membrane.
- Students think pregnancy is mostly about the baby. Maternal physiology changes dramatically: blood volume, cardiac output, renal function, respiratory drive.
- hCG is sometimes thought to come from the mother. It comes from trophoblast (fetal-side) tissue, which is why it is such a specific pregnancy marker.

Order of operations (what to teach first)

Fertilization and implantation → hormonal support (hCG, corpus luteum, placenta) → placental functions → maternal physiology by trimester.

Self-test prompts

- Where does fertilization typically occur?
- What is the role of hCG in early pregnancy?
- Why does maternal blood volume rise during pregnancy?

Why This Matters to Your Patients

Ectopic pregnancy: implantation outside the uterus, usually in the tube. Rupture produces hemorrhagic shock; this is a surgical emergency. Preeclampsia: third-trimester hypertension and proteinuria, with risk of progression to eclampsia (seizures); blood pressure monitoring and timely delivery are the management. Gestational diabetes: places mother at risk of preeclampsia and fetus at risk of macrosomia and shoulder dystocia at delivery. Supine hypotensive syndrome: a third-trimester patient becomes dizzy lying flat. Roll her left to relieve IVC compression. These are the prenatal nursing scenarios this unit prepares students for.